



Movie Recommender

Web Application

Capstone-I Project | Group No. 191 | IIT Patna,
Bihta - 801106

Guided by: Mr. Kuldeep Sir

Team: Aman Vishwakarma | Vishwa Patel | Vaikunth Verma | Ramanuj
Vishwakarma

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Overview of the Movie Recommender Application

A COMPREHENSIVE FULL-STACK SOLUTION

This full-stack Movie Recommender Web Application enables users to search for movies and receive personalized recommendations based on content, utilizing advanced machine learning algorithms for optimal user experience.

System Architecture Overview

API COMMUNICATION

The React frontend communicates with the Flask backend through clearly defined REST APIs. This architecture allows for smooth data exchange and enables the application to function dynamically, responding to user inputs effectively.

RECOMMENDATION INTEGRATION

Content-based recommendations leverage the TMDB API for live movie data. By using TF-IDF vectorization and Cosine Similarity, the system provides personalized suggestions, enhancing user experience and engagement through relevant movie options.

Team Contributions by Aman Vishwakarma

BACKEND DEVELOPMENT

Aman built and configured the Flask backend server with **robust REST API** routing, ensuring that all functionalities were seamlessly integrated and accessible.

API INTEGRATION

He developed essential API endpoints, including apisearch, apirecommend, and apimovie, and integrated the TF-IDF Cosine Similarity recommendation engine for optimized performance.

Team Contributions by Vishwa Patel

FRONTEND DEVELOPMENT

Vishwa developed a complete multi-page React Vite application, enabling seamless user experiences across various sections such as Home, Search, Movie Details, and Recommendations.

API INTEGRATION

He successfully integrated the TMDB API to fetch live movie posters, ratings, and descriptions, enhancing the user interface with dynamic content and improved engagement.

Team Contributions by Vaikunth Verma

UI COMPONENT DEVELOPMENT

Developed reusable UI components like movie cards and navigation bars, ensuring a cohesive design language across the application. This approach helped maintain consistency and enhanced user experience throughout all pages.

DESIGN CONSISTENCY

Contributed significantly to layout decisions and design consistency across various screens. Assisted in frontend debugging efforts, ensuring smooth functionality and a polished final product for user interaction.

Team Contributions by Ramanuj Vishwakarma

DATA SOURCING

Sourced and collected the complete movie dataset, including essential titles, genres, ratings, and descriptions, ensuring a rich and diverse data foundation for our application.

DATA CLEANING

Cleaned the raw data by handling missing values, removing duplicates, and normalizing formats to prepare the dataset for efficient processing in the recommendation model.

Key Features of the Application

MOVIE SEARCH

Users can search for any movie by title and receive **instant live results**, ensuring a seamless experience in discovering films relevant to their interests.

SMART RECOMMENDATIONS

The application utilizes content-based filtering to provide **personalized recommendations**, suggesting similar movies based on users' preferences, including genre, description, and cast details.

DARK MODE TOGGLE

Users have the option to switch between light and dark themes, enhancing their viewing experience while ensuring **comfort and accessibility** during extended usage.

Key Learnings from the Project

FULL-STACK DEVELOPMENT

Gained proficiency in integrating **frontend technologies** like React with backend frameworks such as Flask, enhancing my understanding of the entire web application lifecycle.

API DESIGN PRINCIPLES

Developed skills in REST API design, focusing on **JSON data handling** and efficient communication between the client and server, critical for modern web applications.

MACHINE LEARNING INTEGRATION

Implemented machine learning concepts, specifically **TF-IDF and Cosine Similarity**, to enhance the recommendation system, providing valuable insights into practical applications of data science.

Future Scope of the Application

USER AUTHENTICATION

Implementing login and signup features will enhance user experience by allowing personalized watchlists and tailored recommendations based on individual preferences and viewing history.

COLLABORATIVE FILTERING

Introducing a hybrid recommendation model using user ratings will improve the accuracy of suggestions, ensuring that users receive content aligned with their tastes and preferences.

EXPANDED DATASET

Incorporating a broader dataset, including international and regional films and web series, will diversify the recommendations and provide users with a wider range of content to explore.

Conclusion: Project Successfully Completed

OUR MOVIE RECOMMENDER APPLICATION IS READY

We have achieved all project goals, ensuring the application functions seamlessly from the user interface to the backend. The project is thoroughly tested and prepared for submission.

Thank You!

Guided by: Mr. Kuldeep Sir

Aman Vishwakarma — Backend Developer

GROUP 191 — TEAM MEMBERS

It was a great learning experience building this together.
Vishwa Patel — Frontend Developer (UI/UX & Core Integration)

Vaikunth Verma — Frontend Developer (UI Components)

Ramanuj Vishwakarma — Data Collection & Preprocessing

